DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING PROJECT REPORT

(Project Semester August-December 2020)

Indian Crime Analysis Using Python

Submitted by

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CERTIFICATE

This is to certify that Varnika Sharma bearing Registration no. 12405288 has completed INT-375 project titled, **"Indian Crime Analysis using Python"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 23-04-2026

DECLARATION

I, Varnika Sharma, student of B.Tech CSE under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 04-04-2026 Signature

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Contents

1. Introduction 8

2. Objectives/ Scope of Analysis 10

2.1 Crime Overview Analysis 10

2.2 Crime Trend Analysis 10

2.3 State-wise Crime Comparison 11

2.4 Category-wise Crime Analysis 11

2.5 Relationship Analysis between Crime Categories 11

2.6 Growth Rate Analysis 11

2.7 Distribution and Statistical Analysis 11

2.8 Correlation Analysis 12

2.9 Scope of the Analysis 12

*3.*Source of Dataset 12

3.1 Data Sources Website Links 13

3.2 Dataset Size and Coverage 13

3.3 Dataset Composition 14

3.4 Data Integration 14

3.5 Additional Resources 15

4. ETL Process 15

4.1 Data Extraction 15

4.2 Data Transformation 18

4.2.1 Data Cleaning 18

4.2.2 Data Integration 18

4.2.3 Feature Engineering 19

4.2.4 Data Aggregation 21

4.3 Data Loading 21

4.4 Summary of ETL Process 24

5. Analysis on Dataset 24

5.1 Crime Overview and Distribution Analysis 24

i. Introduction 24

ii. General Description 25

iii. Specific Requirements, Functions and Formulas 25

iv. Analysis Results 25

v. Visualization 25

5.2 Crime Trend Analysis 26

i. Introduction 26

ii. General Description 26

iii. Specific Requirements, Functions and Formulas 26

iv. Analysis Results 26

v. Visualization 26

5.3 State-wise Crime Analysis 27

i. Introduction 27

ii. General Description 27

iii. Specific Requirements, Functions and Formulas 27

iv. Analysis Results 27

v. Visualization 27

5.4 Category-wise Crime Analysis 28

i. Introduction 28

ii. General Description 28

iii. Specific Requirements, Functions and Formulas 28

iv. Analysis Results 28

v. Visualization 28

5.5 Relationship Analysis between Crime Categories 29

i. Introduction 29

ii. General Description 29

iii. Specific Requirements, Functions and Formulas 29

iv. Analysis Results 29

v. Visualization 29

5.6 Growth Rate Analysis 30

i. Introduction 30

ii. General Description 30

iii. Specific Requirements, Functions and Formulas 30

iv. Analysis Results 30

v. Visualization 30

5.7 Distribution and Statistical Analysis 31

i. Introduction 31

ii. General Description 31

iii. Specific Requirements, Functions and Formulas 31

iv. Analysis Results 31

v. Visualization 31

5.8 Correlation Analysis 32

i. Introduction 32

ii. General Description 32

iii. Specific Requirements, Functions and Formulas 32

iv. Analysis Results 32

v. Visualization 32

6. List of Analysis with Results 33

6.1 Introduction 33

6.2 Summary of Analyses and Results 33

6.2.1 Crime Overview and Distribution Analysis 33

6.2.2 Crime Trend Analysis 33

6.2.3 State-wise Crime Analysis 34

6.2.4 Category-wise Crime Analysis 34

6.2.5 Relationship Analysis between Crime Categories 34

6.2.6 Growth Rate Analysis 34

6.2.7 Distribution and Statistical Analysis 34

6.2.8 Correlation Analysis 34

6.3 Overall Insights 35

6.4 Conclusion of Analysis 35

7. References 35

8. Bibliography 36

9. Project Links 36

1. Introduction

Crime is one of the most significant social issues affecting the safety, stability, and development of any nation. It not only impacts individuals and communities but also influences economic growth, governance, and public trust in institutions. In a diverse and densely populated country like India, crime patterns vary widely across different states, regions, and socio-economic conditions. Therefore, understanding crime trends and distributions is essential for effective policy-making, law enforcement, and societal awareness.

In recent years, the nature of crime has undergone a substantial transformation due to rapid urbanization, technological advancement, and increased digital connectivity. While traditional crimes such as theft, assault, and homicide continue to persist, modern forms of crime, particularly cybercrime, have shown a significant rise. Additionally, crimes against vulnerable sections of society, such as women and children, remain a major concern and require focused attention. These evolving patterns highlight the need for comprehensive and multi-dimensional analysis of crime data.

With the increasing availability of large-scale datasets, data science and analytics have emerged as powerful tools for understanding complex patterns within crime data. Traditional methods of analysis often fail to capture hidden relationships and trends, especially when dealing with high-dimensional datasets. Data visualization and interactive dashboards provide an effective solution by transforming raw data into meaningful insights that are easy to interpret and explore.

This project focuses on the analysis and visualization of crime data across India using Python-based tools and libraries. The primary objective is to convert raw and unstructured crime datasets into an interactive dashboard that enables users to explore data dynamically. The dashboard provides a unified platform for analyzing crime patterns across multiple dimensions, including time, geography, and crime categories.

The datasets used in this project include:

- IPC (Indian Penal Code) crimes
- Crimes against women
- Crimes against children
- Cyber crimes

These datasets are integrated to provide a comprehensive overview of crime patterns across different states and years. By combining multiple data sources, the project ensures a holistic analysis rather than a fragmented view.

The developed dashboard enables users to perform the following tasks:

- Analyze overall crime statistics across India
- Study year-wise crime trends and growth patterns
- Compare crime levels across different states
- Examine category-wise crime distribution
- Identify high-crime regions and emerging trends
- Explore relationships between different crime categories

One of the key aspects of this project is the use of interactive filters and slicers, which allow users to customize their analysis based on specific requirements. Users can select particular states, years, or crime categories and observe how the visualizations change dynamically. This interactivity enhances the usability of the dashboard and makes it a powerful decision-support tool.

From a technical perspective, the project leverages Python libraries such as Pandas and NumPy for data processing, hvPlot and Plotly for visualization, and Panel for dashboard development. These tools enable efficient handling of large datasets and the creation of visually appealing and interactive charts.

In addition to descriptive analysis, the project also incorporates statistical and analytical techniques such as growth rate calculation, correlation analysis, and distribution analysis. These techniques help in identifying patterns, relationships, and anomalies within the data, providing deeper insights into crime trends.

The importance of this project lies in its ability to demonstrate how data-driven approaches can be used to address real-world problems. By transforming complex datasets into interactive visualizations, the dashboard simplifies the process of understanding crime patterns and supports informed decision-making.

Furthermore, this project highlights the role of data visualization in bridging the gap between raw data and actionable insights. It shows how analytical tools can be used not only for data exploration but also for communicating findings effectively to a broader audience.

Overall, the Crime Analysis Dashboard serves as a comprehensive platform for analyzing crime data in India. It combines data integration, preprocessing, visualization, and interactivity to provide meaningful insights into crime patterns. The project demonstrates the practical application of data science techniques in solving real-world problems and emphasizes the importance of data-driven analysis in modern decision-making systems.

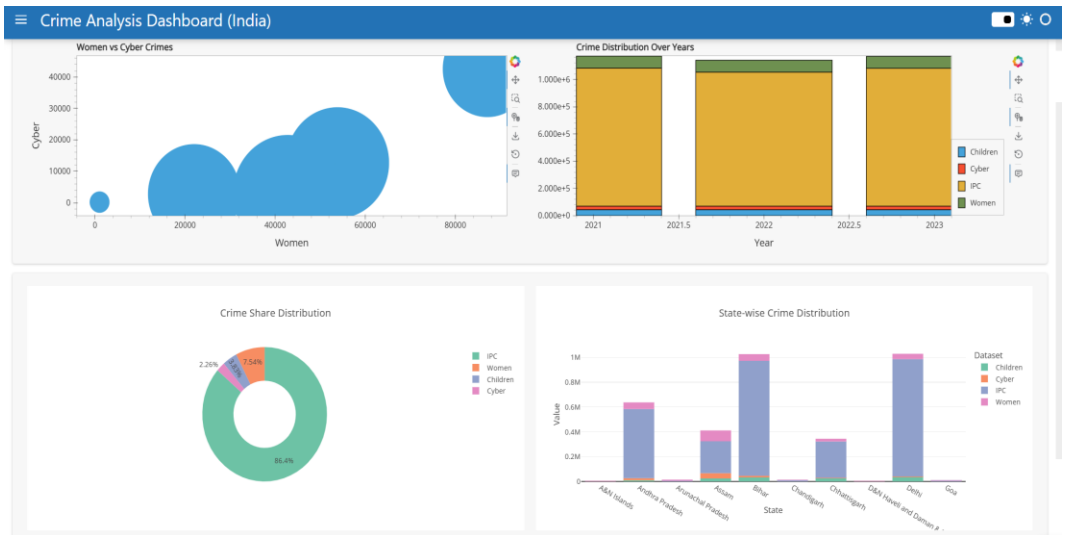


Figure 1: Overview of the Interactive Crime Analysis Dashboard developed using Python

2. Objectives/ Scope of Analysis

The primary objective of this project is to analyze crime data across India using data science techniques and to extract meaningful insights through visualization and statistical analysis. The scope of this analysis is carefully defined to ensure that each objective is supported by corresponding visualizations and analytical outputs.

The objectives of the project are as follows:

2.1 Crime Overview Analysis

The first objective is to provide a comprehensive overview of crime statistics across India. This includes analyzing the total number of crimes and understanding how different crime categories contribute to the overall crime distribution.

This objective helps in establishing a foundational understanding of the dataset by summarizing key metrics and presenting them in a simplified manner. It allows identification of dominant crime categories and provides a baseline for further analysis.

2.2 Crime Trend Analysis

This objective focuses on analyzing how crime patterns change over time. It involves studying year-wise variations in crime data to identify trends, fluctuations, and overall growth patterns.

By examining temporal changes, this analysis helps in understanding whether crime levels are increasing, decreasing, or remaining stable. It also provides insights into long-term patterns and possible future directions.

2.3 State-wise Crime Comparison

The third objective is to compare crime levels across different states and union territories. Since crime distribution varies geographically, this analysis helps in identifying regions with high and low crime rates.

This objective enables the identification of top contributing states and highlights regional disparities in crime patterns, providing a spatial perspective to the analysis.

2.4 Category-wise Crime Analysis

This objective focuses on analyzing crime data based on different categories, specifically crimes against women, children, and cyber crimes.

It aims to compare these categories, understand their contribution to total crime, and analyze how each category behaves over time. This helps in identifying which categories are more prominent and which are emerging as significant concerns.

2.5 Relationship Analysis between Crime Categories

This objective involves analyzing the relationship between different crime categories using comparative techniques.

By examining how crimes against women, children, and cyber crimes vary relative to each other, this analysis helps in identifying patterns and dependencies between categories. It provides insights into whether increases in one category are associated with changes in others.

2.6 Growth Rate Analysis

The purpose of this objective is to measure the rate of change in crime over time.

This involves calculating year-over-year growth rates to identify periods of increase or decrease in crime levels. Growth rate analysis provides a quantitative measure of change and helps in identifying significant shifts in crime patterns.

2.7 Distribution and Statistical Analysis

This objective focuses on understanding the distribution of crime data using statistical visualization techniques.

It includes analyzing the spread of data, identifying outliers, and examining whether the data is skewed. Techniques such as box plots and histograms are used to study the distribution and variability of crime values.

This analysis helps in understanding the nature of the dataset and ensures that insights are not biased by extreme values.

2.8 Correlation Analysis

The final objective is to analyze relationships between different quantitative variables using correlation techniques.

This includes studying how variables such as crime value, growth rate, and contribution percentage are related to each other. Correlation analysis helps in identifying patterns and dependencies within the data, providing deeper analytical insights.

2.9 Scope of the Analysis

The scope of this project is limited to the analysis of crime data across India based on the available datasets. The study focuses on major crime categories and covers a defined time period.

The analysis is primarily descriptive and exploratory in nature, supported by statistical techniques and visualization methods. Advanced predictive modeling and machine learning techniques are not included within the scope of this project.

The findings are based on the processed dataset and the implemented visualizations, and they aim to provide meaningful insights into crime patterns without making predictive assumptions.

3.Source of Dataset

The quality and reliability of any data analysis project largely depend on the source and structure of the dataset used. For this project, crime data was collected from authentic and publicly available government sources to ensure accuracy, credibility, and consistency in analysis.

The datasets used in this study provide comprehensive information about crime statistics across India, covering multiple categories, states, and time periods. These datasets form the foundation for all subsequent data processing, analysis, and visualization.

3.1 Data Sources Website Links

The datasets used in this project were obtained from reliable and publicly available government sources to ensure accuracy, authenticity, and consistency of the analysis.

The following datasets were utilized:

IPC Crimes State/UT-wise Dataset
Source: National Crime Records Bureau (NCRB) – Crime in India Reports

Crimes Against Women Dataset
Source: National Crime Records Bureau (NCRB)

Crimes Against Children Dataset
Source: National Crime Records Bureau (NCRB)

Cyber Crimes Dataset
Source: National Crime Records Bureau (NCRB)

These datasets provide comprehensive crime-related information categorized by state, year, and crime type, enabling detailed analysis and visualization.

3.2 Dataset Size and Coverage

The dataset used in this project covers multiple states and union territories across India and spans a defined time period of analysis.

Key characteristics of the dataset include:

- Coverage of major states and union territories of India
- Year-wise data enabling temporal analysis
- Category-wise classification of crimes
- Multiple datasets combined to provide a comprehensive view

The data spans across multiple years, allowing the analysis of trends and growth patterns over time. The inclusion of various crime categories ensures that the dataset captures different dimensions of crime, making the analysis more comprehensive.

The dataset contains a large number of records, which provides sufficient data for meaningful statistical analysis and visualization. This wide coverage enhances the reliability of insights derived from the project.

3.3 Dataset Composition

The datasets used in this project are structured in a tabular format, where each row represents a specific observation defined by attributes such as state, year, and crime category.

The key attributes present in the dataset include:

• **State** – Represents the name of the state or union territory
• **Year** – Indicates the year of recorded data
• **Dataset** – Represents the category of crime (e.g., Women, Children, Cyber)
• **Value** – Represents the number of reported crime cases

In addition to these core attributes, the dataset was further enhanced during preprocessing by creating additional features such as:

• Growth Rate
• Yearly Change
• Contribution Percentage
• Log-transformed values (for distribution analysis)

The structured nature of the dataset enables efficient data manipulation, aggregation, and visualization using Python libraries such as Pandas and NumPy.

3.4 Data Integration

Since the analysis involves multiple datasets representing different categories of crime, data integration plays a crucial role in ensuring consistency and accuracy.

The datasets were integrated based on common attributes, primarily:

• State
• Year

By using these common fields, data from different sources was combined into a unified structure, allowing for multi-dimensional analysis. This integration enables comparisons across different crime categories and ensures that all datasets align properly for visualization.

During integration, care was taken to maintain consistency in naming conventions and data formats. For example, state names were standardized across datasets to avoid mismatches during merging operations.

The integration process ensures that the dataset behaves as a single cohesive unit, allowing seamless filtering and analysis within the dashboard.

3.5 Additional Resources

In addition to the primary datasets, several supporting resources were utilized to enhance the development and analysis process.

These include:

• **Python Libraries Documentation**
– Pandas (data manipulation)
– NumPy (numerical computation)
– hvPlot and Plotly (data visualization)
– Panel (dashboard development)

• **Online Learning Resources and Tutorials**
– Used for understanding dashboard design and visualization best practices

• **Research Articles and Blogs**
– Provided insights into crime data analysis techniques and statistical methods

These resources played a significant role in improving the efficiency, accuracy, and presentation quality of the project. They also contributed to the implementation of best practices in data preprocessing, analysis, and visualization.

4. ETL Process

The ETL (Extract, Transform, Load) process is a fundamental step in any data analysis project. It involves collecting raw data, cleaning and transforming it into a usable format, and preparing it for analysis and visualization.

In this project, the ETL process was implemented using Python libraries such as Pandas and NumPy. The objective was to convert raw crime datasets into a structured and analysis-ready format that could be used for generating insights through visualization and statistical analysis.

The ETL process consists of three main stages: Extraction, Transformation, and Loading.

4.1 Data Extraction

The extraction phase involves collecting raw data from various sources. In this project, multiple datasets related to crime statistics were obtained from government platforms such as the National Crime Records Bureau (NCRB) and the Open Government Data platform.

The extracted datasets included:

• IPC (Indian Penal Code) crimes

• Crimes against women

• Crimes against children

• Cyber crimes

These datasets were typically available in CSV format and contained information such as state, year, crime category, and number of cases.

The datasets were loaded into the Python environment using Pandas, which allows efficient handling of structured data.

Example:

```
[1]: # ==============================
     # STEP 1: Import All Libraries
     # ==============================

     # Data handling
     import pandas as pd
     import numpy as np

     # Visualization (EDA)
     import matplotlib.pyplot as plt
     import seaborn as sns

     # Interactive visualization (Dashboard)
     import plotly.express as px
     import plotly.io as pio

     # Set plot styles
     sns.set(style="whitegrid")
     pio.renderers.default = "notebook"

     print("All libraries imported successfully!")

     All libraries imported successfully!

[2]: # ==============================
     # STEP 2: Define Paths
     # ==============================

     base_path = "C:/Users/Varnika Sharma/Desktop/Crime Analysis/Crime Analysis/01_Data/Cleaned_Dataset/"

     ipc_path = base_path + "IPC_CRIMES/"
     women_path = base_path + "WOMEN_CRIMES/"
     children_path = base_path + "CHILDREN_CRIMES/"
     cyber_path = base_path + "CYBER_CRIMES/"

     print("Paths set successfully!")

     Paths set successfully!
```

Figure 2: Loading Dataset using Pandas

```
[95]: # ==============================
# STEP 3: Universal Processing Function
# ==============================

def process_dataset(file_path, year, dataset_name):
    df = pd.read_excel(file_path)

    # Clean column names
    df.columns = df.columns.str.strip()

    # Handle State column variations
    if "State/UT" in df.columns:
        df.rename(columns={"State/UT": "State"}, inplace=True)
    elif "STATE/UT" in df.columns:
        df.rename(columns={"STATE/UT": "State"}, inplace=True)
    elif "State" not in df.columns:
        df.rename(columns={df.columns[1]: "State"}, inplace=True)

    # Drop serial column if exists
    if "Sl" in df.columns:
        df.drop(columns=["Sl"], inplace=True)

    # Convert wide → long
    df_long = pd.melt(
        df,
        id_vars=["State"],
        var_name="Crime_Info",
        value_name="Value"
    )

    # Split Crime_Info into parts
    df_long[["Category", "Crime", "Metric"]] = df_long["Crime_Info"].str.rsplit("_", n=2, expand=True)
    # Clean values
    df_long["Value"] = pd.to_numeric(df_long["Value"], errors="coerce")
    df_long.dropna(inplace=True)


    # Remove totals (very important)
    df_long = df_long[~df_long["Crime"].str.contains("total|all", case=False, na=False)]

    # Add metadata
    df_long["Year"] = year
    df_long["Dataset"] = dataset_name
    df_long.loc[df_long["Dataset"].str.contains("Cyber", case=False, na=False), "Metric"] = "I"

    print(f"{dataset_name} {year} processed successfully. Shape:", df_long.shape)

    return df_long
```

Figure 3: Loading Dataset using Pandas

```
[96]: # ==============================
# STEP 4: Load Datasets (2021)
# ==============================

ipc_2021 = process_dataset(ipc_path + "IPC_Crimes_Head-wise_Raw_2021.xlsx", 2021, "IPC")
women_2021 = process_dataset(women_path + "Crimes_Against_Women_Head-wise_State-UT_Raw_2021.xlsx", 2021, "Women")
children_2021 = process_dataset(children_path + "Crimes_Against_Children_State-UT_2021.xlsx", 2021, "Children")
cyber_2021 = process_dataset(cyber_path + "Cyber_Crimes_State-UT_2021.xlsx", 2021, "Cyber")

IPC 2021 processed successfully. Shape: (12816, 8)
Women 2021 processed successfully. Shape: (4572, 8)
Children 2021 processed successfully. Shape: (5976, 8)
Cyber 2021 processed successfully. Shape: (1548, 8)

[97]: # ==============================
# STEP 5: Combine Datasets
# ==============================

final_df = pd.concat(
    [ipc_2021, women_2021, children_2021, cyber_2021],
    ignore_index=True
)

print("Final dataset shape:", final_df.shape)
display(final_df.head())

Final dataset shape: (24912, 8)
```

	State	Crime_Info	Value	Category	Crime	Metric	Year	Dataset
0	Andhra Pradesh	Offences affecting the Human Body_Murder (Sec....	956.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
1	Arunachal Pradesh	Offences affecting the Human Body_Murder (Sec....	49.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
2	Assam	Offences affecting the Human Body_Murder (Sec....	1192.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
3	Bihar	Offences affecting the Human Body_Murder (Sec....	2799.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
4	Chhattisgarh	Offences affecting the Human Body_Murder (Sec....	1007.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC

Figure 4: Loading Dataset using Pandas

4.2 Data Transformation

The transformation phase is the most critical part of the ETL process. It involves cleaning, structuring, and enhancing the data to make it suitable for analysis.

In this project, several transformation steps were performed:

4.2.1 Data Cleaning

Raw datasets often contain inconsistencies such as missing values, incorrect data types, and duplicate records.

The following steps were performed:

- Removal of null or missing values
- Conversion of data types (e.g., Year to integer, Value to numeric)
- Standardization of column names
- Removal of duplicate entries

Example:

```
[103]: # ============================
       # STEP 9: Final Cleaning
       # ============================

       # Keep only actual cases
       df_clean = final_df[final_df["Metric"] == "I"].copy()

       # Remove unwanted aggregated entries
       df_clean = df_clean[~df_clean["Crime"].str.contains("total|all", case=False, na=False)]

       # Convert to numeric again (safe)
       df_clean["Value"] = pd.to_numeric(df_clean["Value"], errors="coerce")

       df_clean.dropna(inplace=True)

       print("Cleaned shape:", df_clean.shape)
       display(df_clean.head())

       Cleaned shape: (27756, 8)
```

	State	Crime_Info	Value	Category	Crime	Metric	Year	Dataset
0	Andhra Pradesh	Offences affecting the Human Body_Murder (Sec....	956.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
1	Arunachal Pradesh	Offences affecting the Human Body_Murder (Sec....	49.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
2	Assam	Offences affecting the Human Body_Murder (Sec....	1192.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
3	Bihar	Offences affecting the Human Body_Murder (Sec....	2799.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
4	Chhattisgarh	Offences affecting the Human Body_Murder (Sec....	1007.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC

Figure 5: Data Cleaning Process

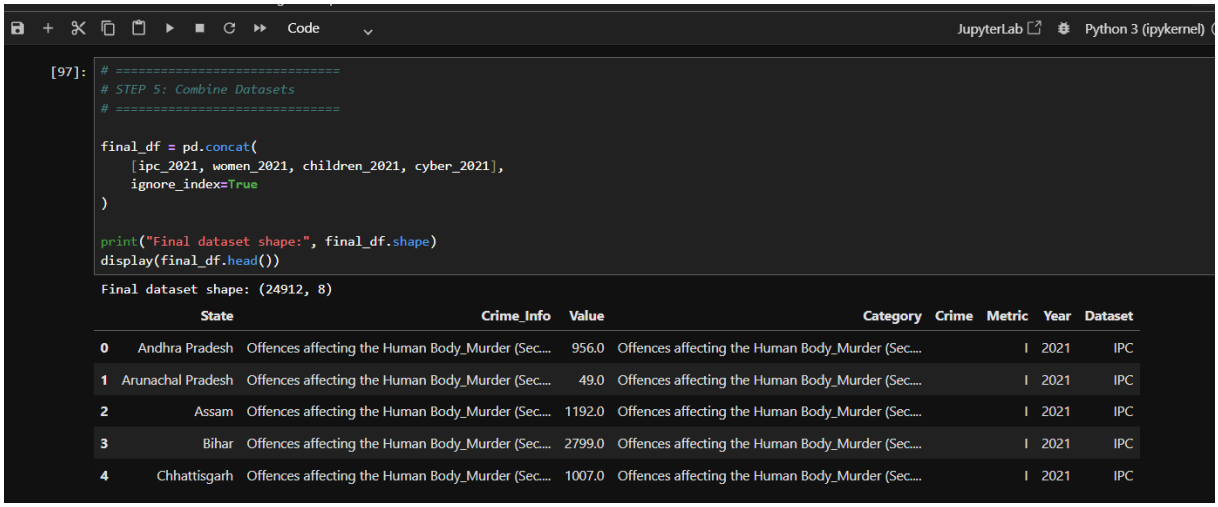
4.2.2 Data Integration

Since multiple datasets were used, they needed to be combined into a unified structure.

This was done using common fields such as:

- State
- Year

Data integration ensures that all datasets align correctly, enabling multi-dimensional analysis across categories.

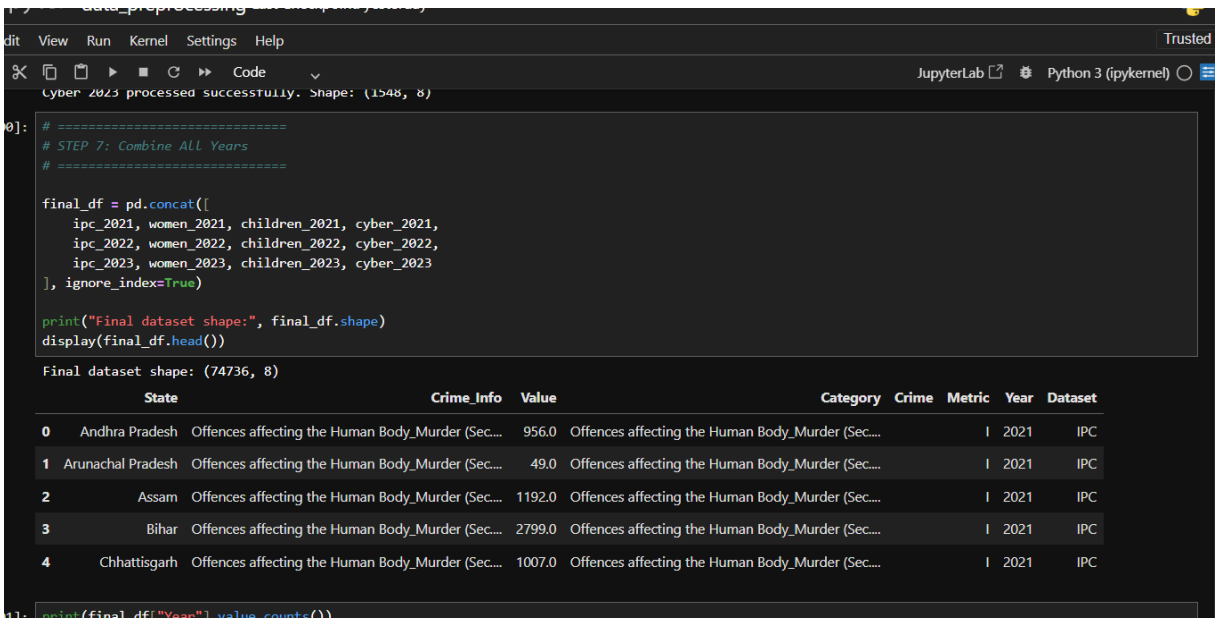


```
[97]: # ==============================
      # STEP 5: Combine Datasets
      # ==============================

      final_df = pd.concat(
          [ipc_2021, women_2021, children_2021, cyber_2021],
          ignore_index=True
      )

      print("Final dataset shape:", final_df.shape)
      display(final_df.head())
```

Final dataset shape: (24912, 8)

	State	Crime_Info	Value	Category	Crime	Metric	Year	Dataset
0	Andhra Pradesh	Offences affecting the Human Body_Murder (Sec....	956.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
1	Arunachal Pradesh	Offences affecting the Human Body_Murder (Sec....	49.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
2	Assam	Offences affecting the Human Body_Murder (Sec....	1192.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
3	Bihar	Offences affecting the Human Body_Murder (Sec....	2799.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
4	Chhattisgarh	Offences affecting the Human Body_Murder (Sec....	1007.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC

Figure 6: Combining Datasets



```
# ==============================
# STEP 7: Combine All Years
# ==============================

final_df = pd.concat([
    ipc_2021, women_2021, children_2021, cyber_2021,
    ipc_2022, women_2022, children_2022, cyber_2022,
    ipc_2023, women_2023, children_2023, cyber_2023
], ignore_index=True)

print("Final dataset shape:", final_df.shape)
display(final_df.head())
```

Final dataset shape: (74736, 8)

	State	Crime_Info	Value	Category	Crime	Metric	Year	Dataset
0	Andhra Pradesh	Offences affecting the Human Body_Murder (Sec....	956.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
1	Arunachal Pradesh	Offences affecting the Human Body_Murder (Sec....	49.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
2	Assam	Offences affecting the Human Body_Murder (Sec....	1192.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
3	Bihar	Offences affecting the Human Body_Murder (Sec....	2799.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC
4	Chhattisgarh	Offences affecting the Human Body_Murder (Sec....	1007.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC

Figure 7: Combined Datasets

4.2.3 Feature Engineering

To enhance the analytical capability of the dataset, additional features were created.

These include:

- **Yearly Change** – Difference in crime values between consecutive years
- **Growth Rate** – Percentage change in crime over time
- **Contribution Percentage** – Share of a category in total crime
- **Log Value** – Used for distribution analysis

Example:

```
[112]: # ==============================
       #  Step-12: Advanced Feature Engineering
       # ==============================

       df_feat = df_clean.copy()

       # 1. Total crimes per state-year
       df_feat["State_Total"] = df_feat.groupby(["State", "Year"])["Value"].transform("sum")

       # 2. Crime share
       df_feat["Crime_Share"] = df_feat["Value"] / df_feat["State_Total"]

       # 3. Dataset total
       df_feat["Dataset_Total"] = df_feat.groupby(["Dataset", "Year"])["Value"].transform("sum")

       # 4. Contribution %
       df_feat["Contribution_%"] = (df_feat["Value"] / df_feat["Dataset_Total"]) * 100

       # 5. Previous year value
       df_feat["Prev_Value"] = df_feat.groupby(["State", "Crime"])["Value"].shift(1)

       # 6. Growth rate
       df_feat["Growth_Rate"] = ((df_feat["Value"] - df_feat["Prev_Value"]) / df_feat["Prev_Value"]) * 100

       # Fill NaN for first year
       df_feat["Growth_Rate"] = df_feat["Growth_Rate"].fillna(0)

       # 7. Rank of crimes within state
       df_feat["Crime_Rank_State"] = df_feat.groupby(["State", "Year"])["Value"].rank(ascending=False)

       # 8. Rank of states (overall)
       df_feat["State_Rank"] = df_feat.groupby("Year")["Value"].rank(ascending=False)

       # 9. High crime flag
       df_feat["High_Crime_Flag"] = df_feat["Value"] > df_feat["Value"].mean()

       # 10. Intensity bucket
       try:
           df_feat["Intensity"] = pd.qcut(df_feat["Value"], q=4, labels=["Low", "Medium", "High", "Very High"], duplicates="drop")
       except:
           df_feat["Intensity"] = pd.cut(df_feat["Value"], bins=3, labels=["Low", "Medium", "High"])
       # 11. Log transformation
       df_feat["Log_Value"] = np.log1p(df_feat["Value"])

       # 12. Z-score (statistical normalization)
       df_feat["Z_Score"] = (df_feat["Value"] - df_feat["Value"].mean()) / df_feat["Value"].std()

       # 13. Yearly change absolute
       df_feat["Yearly_Change"] = df_feat["Value"] - df_feat["Prev_Value"]
```

Figure 8: Feature Engineering Columns

```
# 13. Yearly change absolute
df_feat["Yearly_Change"] = df_feat["Value"] - df_feat["Prev_Value"]
df_feat["Yearly_Change"] = df_feat["Yearly_Change"].fillna(0)

# 14. Dataset share within state
df_feat["Dataset_State_Share"] = df_feat["Value"] / df_feat.groupby(["State","Year"])["Value"].transform("sum")

# 15. Crime frequency category
df_feat["Frequency"] = pd.cut(
    df_feat["Value"],
    bins=3,
    labels=["Rare", "Moderate", "Frequent"]
)

print("Total columns:", len(df_feat.columns))
display(df_feat.head())
```

Total columns: 23

	State	Crime_Info	Value	Category	Crime	Metric	Year	Dataset	State_Total	Crime_Share	...	Growth_Rate	Crime_Rank_State	State_Rank	High_Crin
0	Andhra Pradesh	Offences affecting the Human Body_Murder (Sec....	956.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC	224927.0	0.004250	...	0.0	32.0	693.5	
1	Arunachal Pradesh	Offences affecting the Human Body_Murder (Sec....	49.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC	3603.0	0.013600	...	0.0	21.0	2168.0	
2	Assam	Offences affecting the Human Body_Murder (Sec....	1192.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC	188338.0	0.006329	...	0.0	36.0	602.0	
3	Bihar	Offences affecting the Human Body_Murder (Sec....	2799.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC	305961.0	0.009148	...	0.0	17.0	335.0	
4	Chhattisgarh	Offences affecting the Human Body_Murder (Sec....	1007.0	Offences affecting the Human Body_Murder (Sec....		I	2021	IPC	113598.0	0.008865	...	0.0	26.0	666.5	

5 rows × 23 columns

Figure 9: Feature Engineering Columns

4.2.4 Data Aggregation

Aggregation is performed to summarize data for analysis.

Examples include:

- Total crimes per state
- Total crimes per year
- Category-wise totals

Example:

```
[115]: crime_stats = df_clean.groupby("Crime")["Value"].agg([
           "mean", "median", "std", "var", "skew"
       ]).sort_values(by="mean", ascending=False).head(10)

       print(crime_stats)

                                                          mean  median  \
Crime
A) Voluntarily Causing Simple Hurt (Sec.323 r/w...  8574.425926  1191.5
B) Voluntarily causing hurt by Dangerous weapon...  4925.157407   409.0
B) Other Accidents (other than Hit and Run)         2807.166667   935.5
A) Voluntarily causing grievous hurt (Sec.325 I...  2051.527778   313.5
Kidnapping and Abduction (Sec.363 IPC)              1730.592593   280.0
A) Hit and Run                                      1324.861111   230.5
Fraud (Sec.420 r/w Sec.465,468-471 IPC/IT Act)       985.731481   213.0
A) Missing Children Deemed as Kidnapped              922.712963    65.5
B) Other Kidnapping & Abduction                      807.879630   188.5
A) Missing Children Deemed as\nKidnapped             798.777778    66.5

                                                            std  \
Crime
A) Voluntarily Causing Simple Hurt (Sec.323 r/w...  17851.637178
B) Voluntarily causing hurt by Dangerous weapon...   8834.542192
B) Other Accidents (other than Hit and Run)          3582.036622
A) Voluntarily causing grievous hurt (Sec.325 I...   4096.888891
Kidnapping and Abduction (Sec.363 IPC)               2777.540810
A) Hit and Run                                       2974.167231
Fraud (Sec.420 r/w Sec.465,468-471 IPC/IT Act)       2012.466588
A) Missing Children Deemed as Kidnapped              1710.186747
B) Other Kidnapping & Abduction                      1302.090323
A) Missing Children Deemed as\nKidnapped             1446.916491

                                                            var      skew
Crime
A) Voluntarily Causing Simple Hurt (Sec.323 r/w...  3.186809e+08  2.590809
B) Voluntarily causing hurt by Dangerous weapon...  7.804914e+07  2.221056
B) Other Accidents (other than Hit and Run)         1.283099e+07  1.538209
A) Voluntarily causing grievous hurt (Sec.325 I...  1.678450e+07  2.694570
Kidnapping and Abduction (Sec.363 IPC)              7.714733e+06  1.974081
A) Hit and Run                                      8.845671e+06  4.146853
Fraud (Sec.420 r/w Sec.465,468-471 IPC/IT Act)      4.050022e+06  5.061172
A) Missing Children Deemed as Kidnapped             2.924739e+06  2.304337
B) Other Kidnapping & Abduction                     1.695439e+06  1.811494
A) Missing Children Deemed as\nKidnapped            2.093567e+06  2.180129
```

Figure 10: Aggregated Data Output

4.3 Data Loading

The final stage of ETL is loading the processed data into a system where it can be used for visualization and analysis.

In this project, the cleaned and transformed dataset was used directly within the Python environment to build an interactive dashboard using Panel, hvPlot, and Plotly.

The processed data is dynamically connected to the dashboard, allowing real-time updates based on user inputs such as filters and slicers.

This eliminates the need for a separate database and ensures efficient data handling within the application.

```python
import hvplot.pandas
import plotly.express as px

pn.extension('plotly')

# ==============================
# LOAD DATA
# ==============================
df = pd.read_csv("C:/Users/Varnika Sharma/Desktop/Crime Analysis/Crime Analysis/01_Data/Final_Dataset/cleaned_crime_data.csv")   # <-- UPDATE PATH

# ==============================
# FILTERS
# ==============================
year_slider = pn.widgets.IntRangeSlider(
    name="Year Range",
    start=int(df["Year"].min()),
    end=int(df["Year"].max()),
    value=(int(df["Year"].min()), int(df["Year"].max()))
)

state_select = pn.widgets.MultiChoice(
    name="Select States",
    options=sorted(df["State"].unique().tolist()),
    value=sorted(df["State"].unique().tolist())[:10]
)

dataset_select = pn.widgets.MultiChoice(
    name="Crime Category",
    options=sorted(df["Dataset"].unique().tolist()),
    value=sorted(df["Dataset"].unique().tolist())
)

# ==============================
# FILTER FUNCTION
# ==============================
def get_data():
    return df[
        (df["Year"].between(*year_slider.value)) &
        (df["State"].isin(state_select.value)) &
        (df["Dataset"].isin(dataset_select.value))
    ]
```

Figure 11: Data Connected to Dashboard

```python
# ==============================
# 1. SCATTER
# ==============================
def scatter_plot():
    data = get_data()

    pivot = (
        data.groupby(["State", "Dataset"])["Value"]
        .sum()
        .unstack(fill_value=0)
        .reset_index()
    )

    return pivot.hvplot.scatter(
        x="Women",
        y="Cyber",
        size="Children",
        hover_cols=["State"],
        height=350,
        title="Women vs Cyber Crimes"
    )

# ==============================
# 2. STACKED BAR
# ==============================
def stacked_bar():
    data = get_data()

    grouped = (
        data.groupby(["Year", "Dataset"])["Value"]
        .sum()
        .unstack()
        .fillna(0)
    )

    return grouped.hvplot.bar(
        stacked=True,
        height=350,
        title="Crime Distribution Over Years"
    )
```

Figure 12: Dataset Connected to Charts

```python
# ==============================
# 3. PIE (COLORED)
# ==============================
def pie_chart():
    data = get_data()

    pie_df = (
        data.groupby("Dataset")["Value"]
        .sum()
        .reset_index()
        .sort_values("Value", ascending=False)
        .head(6)
    )

    fig = px.pie(
        pie_df,
        names="Dataset",
        values="Value",
        hole=0.5,
        title="Crime Share Distribution",
        color="Dataset",
        color_discrete_sequence=px.colors.qualitative.Set2
    )

    return pn.pane.Plotly(fig, height=400)

# ==============================
# 4. STATE BAR (COLORED)
# ==============================
def state_bar():
    data = get_data()

    state_df = (
        data.groupby(["State", "Dataset"])["Value"]
        .sum()
        .reset_index()
    )

    fig = px.bar(
        state_df,
        x="State",
        y="Value",
        color="Dataset",
        title="State-wise Crime Distribution",
        color_discrete_sequence=px.colors.qualitative.Set2
    )

    return pn.pane.Plotly(fig, height=400)
```

Figure 13: Dataset Connected to Charts

```python
# ==============================
# 5. TREND LINE (COLORED)
# ==============================
def trend_line():
    data = get_data()

    trend_df = (
        data.groupby("Year")["Value"]
        .sum()
        .reset_index()
    )

    fig = px.line(
        trend_df,
        x="Year",
        y="Value",
        markers=True,
        title="Year-wise Crime Trend",
        color_discrete_sequence=["#2a9d8f"]   # clean teal
    )

    return pn.pane.Plotly(fig, height=400)

# ==============================
# 6. GROWTH RATE (COLORED)
# ==============================
def growth_chart():
    data = get_data()

    df_year = (
        data.groupby("Year")["Value"]
        .sum()
        .reset_index()
    )

    df_year["Growth"] = df_year["Value"].pct_change() * 100
    df_year = df_year.dropna()

    fig = px.bar(
        df_year,
        x="Year",
        y="Growth",
        title="Year-wise Crime Growth Rate (%)",
        color="Growth",
        color_continuous_scale="RdYlGn"
    )

    return pn.pane.Plotly(fig, height=400)
```

Figure 14: Dataset Connected to Carts

```python
# ==============================
# BIND
# ==============================
scatter_p = pn.bind(scatter_plot)
bar_p = pn.bind(stacked_bar)
pie_p = pn.bind(pie_chart)
state_p = pn.bind(state_bar)
trend_p = pn.bind(trend_line)
growth_p = pn.bind(growth_chart)

# ==============================
# DASHBOARD
# ==============================
dashboard = pn.template.FastListTemplate(
    title="Crime Analysis Dashboard (India)",

    sidebar=[
        pn.pane.Markdown("## Filters"),
        year_slider,
        state_select,
        dataset_select
    ],

    main=[
        pn.pane.Markdown("## Overview"),

        pn.Row(scatter_p, bar_p),

        pn.Row(pie_p, state_p),

        pn.Row(trend_p, growth_p)
    ]
)

# ==============================
# RUN
# ==============================
dashboard.show()
```

Figure 15: Dataset Connected to Charts

4.4 Summary of ETL Process

The ETL process ensures that raw data is converted into a structured and meaningful format suitable for analysis.

• Extraction collects raw datasets from reliable sources
• Transformation cleans, integrates, and enhances the data
• Loading prepares the data for visualization and interaction

This structured approach ensures data accuracy, consistency, and usability, forming the backbone of the entire analysis process.

5. Analysis on Dataset

5.1 Crime Overview and Distribution Analysis

i. Introduction

Crime distribution analysis provides a high-level understanding of how different categories of crimes contribute to the overall crime scenario. It serves as the starting point for the analysis by summarizing the dataset into meaningful proportions.

ii. General Description

In this analysis, the total number of crimes is grouped based on categories such as crimes against women, children, and cyber crimes. The objective is to understand the relative contribution of each category to the total crime count. This helps in identifying dominant crime categories and areas that require focused attention.

iii. Specific Requirements, Functions and Formulas

The analysis is performed by aggregating the dataset based on the category column. The total crime value for each category is calculated using grouping and summation functions.

Example logic:

```
df.groupby("Dataset")["Value"].sum()
```

iv. Analysis Results

The results indicate that crime distribution is uneven across categories. Certain categories contribute significantly more to the overall crime count, highlighting key areas of concern.

v. Visualization

The analysis is represented using a donut chart, which clearly shows the proportional contribution of each category to the total crime.

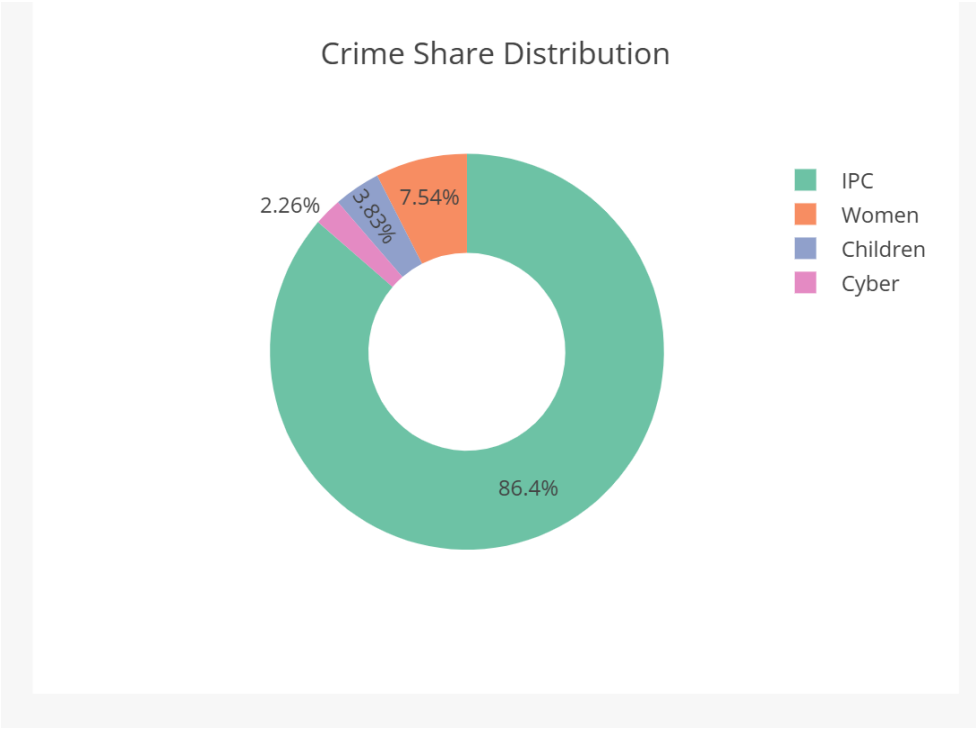


Figure 16: Donut Chart (Crime Distribution)

5.2 Crime Trend Analysis

i. Introduction

Crime trend analysis focuses on understanding how crime patterns change over time. It helps in identifying whether crime levels are increasing, decreasing, or remaining stable.

ii. General Description

This analysis examines year-wise variations in crime data. By analyzing trends across multiple years, it becomes possible to detect long-term patterns and significant fluctuations.

iii. Specific Requirements, Functions and Formulas

The dataset is grouped by year, and total crime values are calculated for each year.

Example logic:

df.groupby("Year")["Value"].sum()

iv. Analysis Results

The results show a clear trend in crime patterns over time. Variations across years indicate changes in crime levels, which may be influenced by social, economic, or technological factors.

v. Visualization

The results are visualized using a line chart, which effectively represents the trend over time.

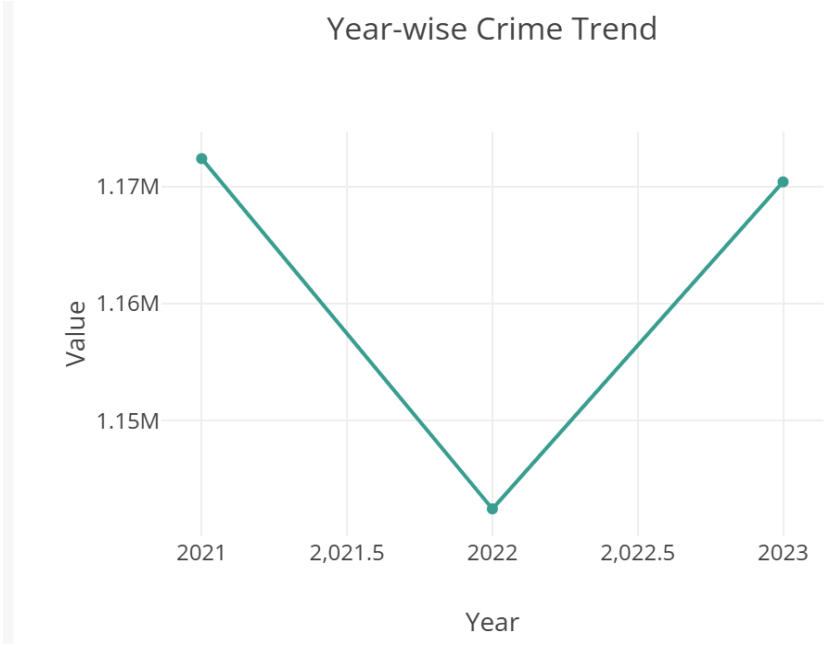


Figure 17: Line Chart (Trend Analysis)

5.3 State-wise Crime Analysis

i. Introduction

State-wise analysis provides insights into the geographical distribution of crime across different regions.

ii. General Description

This analysis compares crime levels across states to identify regions with high and low crime rates. It highlights regional disparities and helps in understanding location-based patterns.

iii. Specific Requirements, Functions and Formulas

The dataset is grouped by state and category to calculate total crime values.

Example logic:

df.groupby(["State", "Dataset"])["Value"].sum()

iv. Analysis Results

The results reveal that crime distribution varies significantly across states. Some states consistently show higher crime levels, while others contribute relatively less.

v. Visualization

A bar chart is used to represent crime distribution across states, allowing easy comparison.

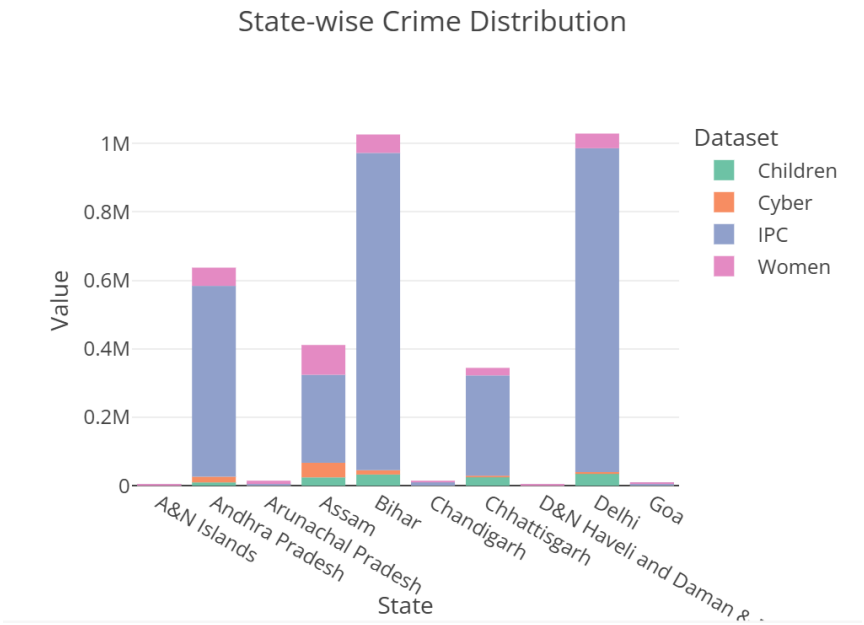


Figure 18: State-Wise Bar Chart

5.4 Category-wise Crime Analysis

i. Introduction

Category-wise analysis focuses on comparing different types of crimes and understanding their contribution to overall crime patterns.

ii. General Description

This analysis examines how different crime categories behave over time and across regions. It provides insights into which categories are more prominent and how they evolve.

iii. Specific Requirements, Functions and Formulas

The dataset is grouped by both year and category to analyze trends across categories.

Example logic:

df.groupby(["Year", "Dataset"])["Value"].sum()

iv. Analysis Results

The results indicate that different categories follow different trends. Some categories show steady growth, while others fluctuate over time.

v. Visualization

A stacked bar chart is used to compare category-wise distribution over years.

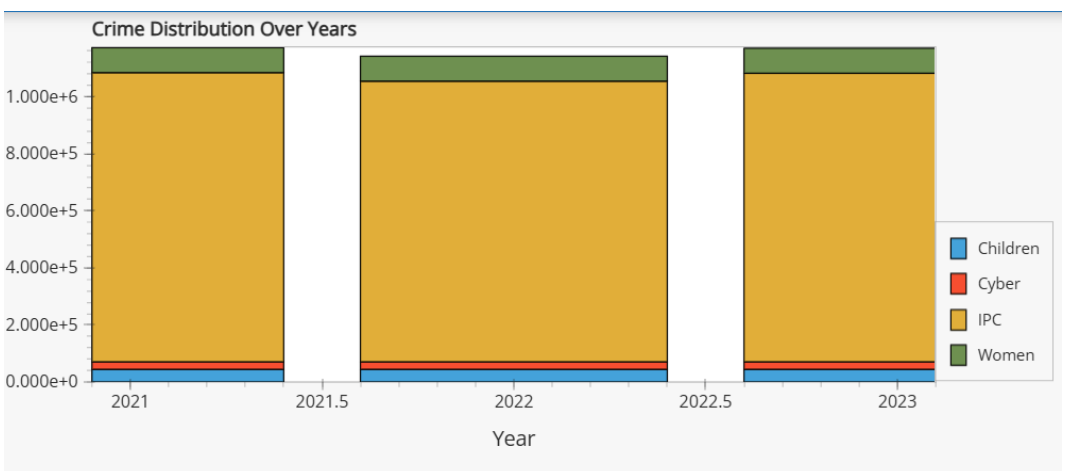


Figure 19: Stacked Bar Chart (Category-Wise)

5.5 Relationship Analysis between Crime Categories

i. Introduction

Relationship analysis helps in understanding how different crime categories are related to each other.

ii. General Description

This analysis compares crimes against women, children, and cyber crimes to identify patterns and dependencies between them.

iii. Specific Requirements, Functions and Formulas

The dataset is transformed into a pivot format where each category becomes a separate column.

Example logic:

df.pivot_table(index="State", columns="Dataset", values="Value")

iv. Analysis Results

The results suggest that there are observable relationships between different crime categories, indicating possible underlying factors influencing multiple types of crimes.

v. Visualization

A scatter plot is used to represent relationships, where different axes correspond to different crime categories.

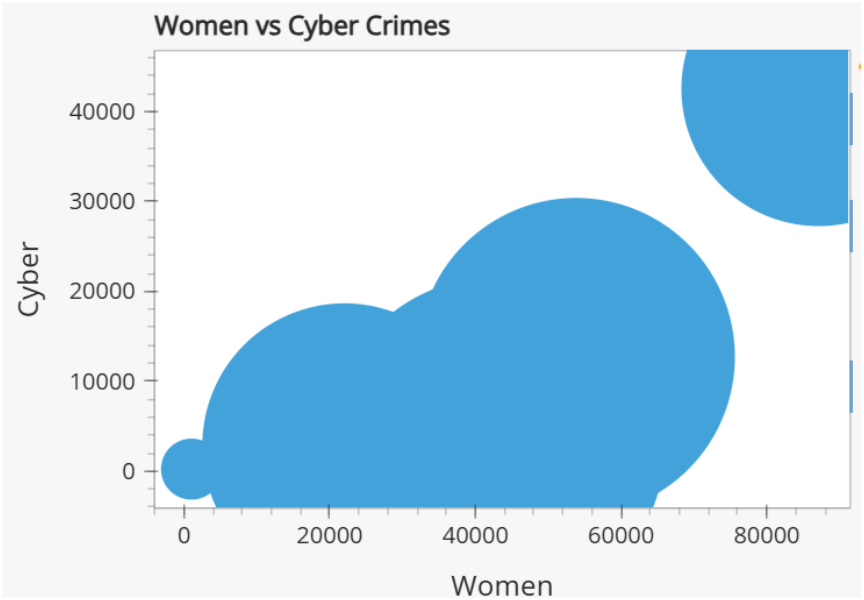


Figure 20: Scatter Plot (Relationship)

5.6 Growth Rate Analysis

i. Introduction

Growth rate analysis measures how crime levels change over time in percentage terms.

ii. General Description

This analysis identifies periods of increase or decrease in crime by calculating the percentage change in values across years.

iii. Specific Requirements, Functions and Formulas

Growth rate is calculated using the percentage change formula:

*df["Growth"] = df["Value"].pct_change() * 100*

iv. Analysis Results

The results highlight years with significant increases or decreases in crime levels, providing insights into major shifts in trends.

v. Visualization

A bar chart is used to display growth rates across years.

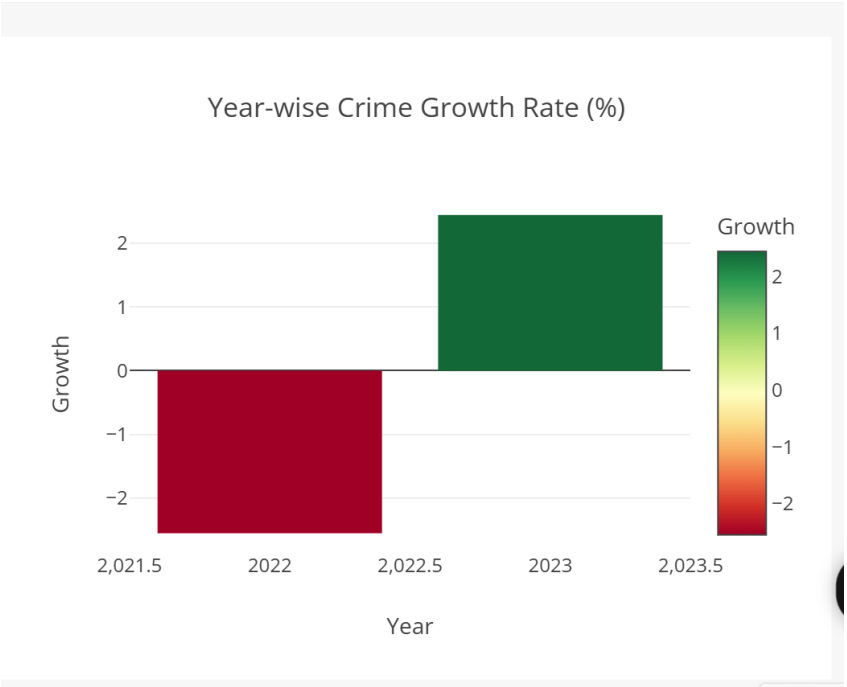


Figure 21: Growth Rate Bar Chart

5.7 Distribution and Statistical Analysis

i. Introduction

Distribution analysis helps in understanding the spread and variability of crime data.

ii. General Description

This analysis focuses on identifying outliers, skewness, and data spread using statistical visualization techniques.

iii. Specific Requirements, Functions and Formulas

Log transformation is applied to normalize skewed data:

np.log1p(df["Value"])

iv. Analysis Results

The analysis reveals that crime data is highly skewed, with a wide range of values. Log transformation helps in better visualization and interpretation.

v. Visualization

Box plots and histograms are used to represent data distribution and identify outliers.

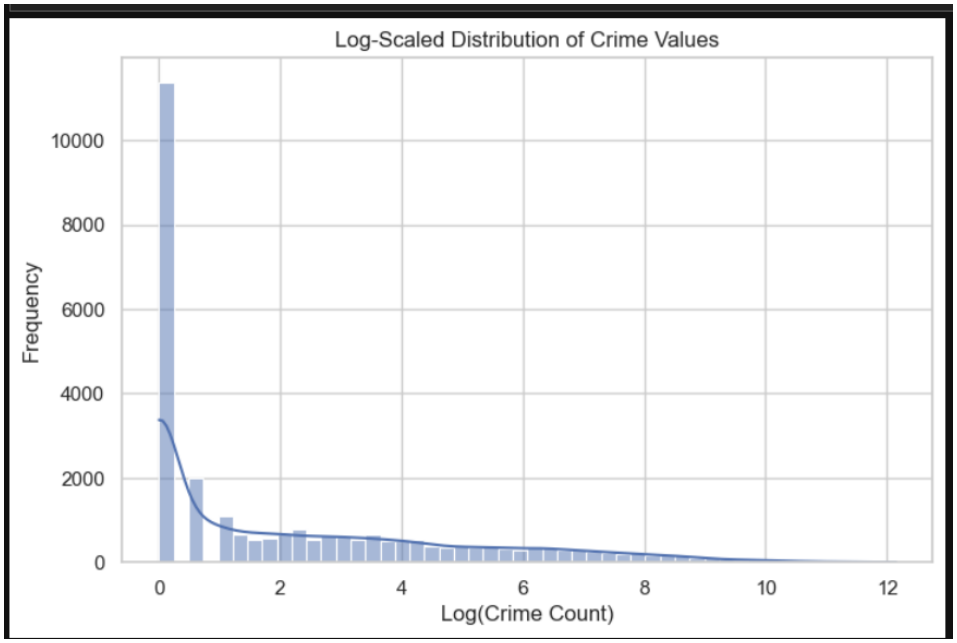


Figure 22: Histogram

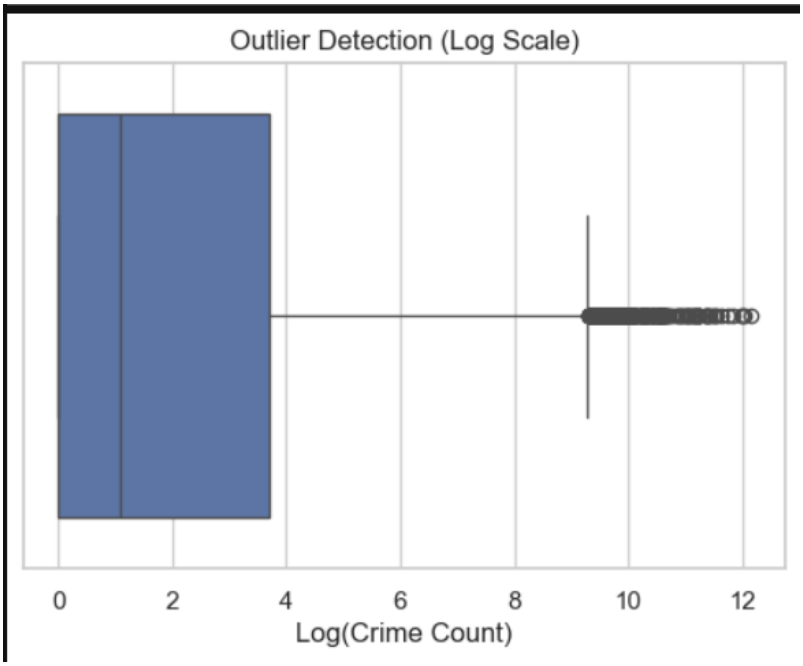


Figure 23: Box Plot

5.8 Correlation Analysis

i. Introduction

Correlation analysis is used to study relationships between different numerical variables in the dataset.

ii. General Description

This analysis examines how variables such as crime value, growth rate, and yearly change are related to each other.

iii. Specific Requirements, Functions and Formulas

Correlation is calculated using:

df.corr()

iv. Analysis Results

The results indicate varying levels of correlation between variables, suggesting relationships and dependencies within the dataset.

v. Visualization

A heatmap is used to represent correlation values visually.

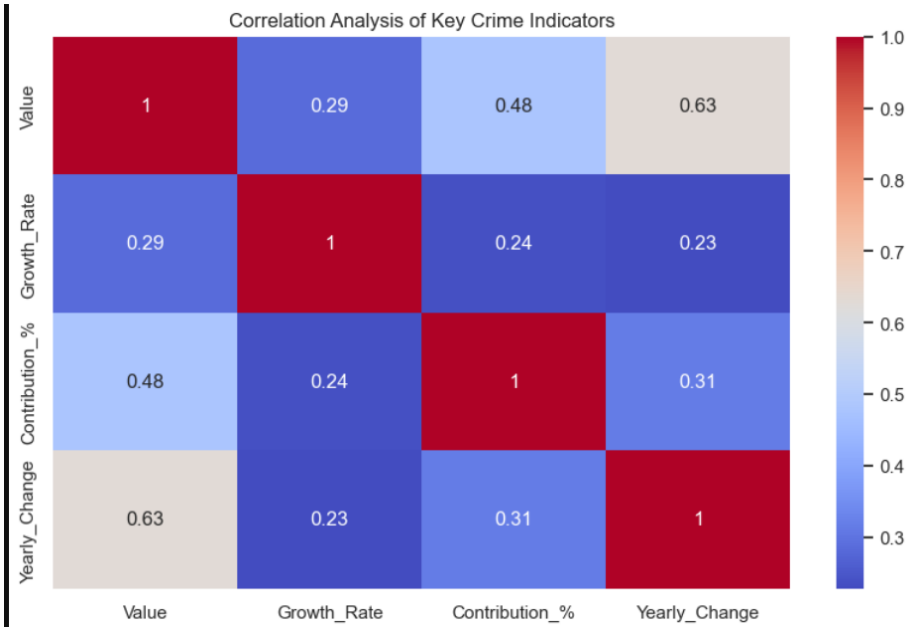


Figure 24: Heatmap

6. List of Analysis with Results

6.1 Introduction

This section provides a summarized view of all the analyses performed in the project along with their key findings. While the previous section presented a detailed explanation of each analysis along with visualizations, this section focuses on highlighting the most important insights derived from the dataset.

The objective of this section is to present the results in a concise and structured manner, making it easier to understand the overall outcomes of the analysis without going into detailed explanations.

6.2 Summary of Analyses and Results

6.2.1 Crime Overview and Distribution Analysis

The analysis of overall crime distribution indicates that crime is not uniformly distributed across categories. Certain categories contribute a significantly larger share to the total crime count, highlighting key areas that require focused attention. This uneven distribution reflects the varying nature and frequency of different types of crimes.

6.2.2 Crime Trend Analysis

The trend analysis shows noticeable variations in crime levels over the years. The overall pattern indicates a gradual change in crime rates, with certain periods showing increases or decreases. These

fluctuations suggest the influence of external factors such as policy changes, population growth, and technological developments.

6.2.3 State-wise Crime Analysis

The state-wise comparison reveals that crime distribution varies significantly across different states. Some states consistently record higher crime levels, while others contribute relatively less. This highlights regional disparities and indicates that crime patterns are influenced by geographic and socio-economic factors.

6.2.4 Category-wise Crime Analysis

The category-wise analysis shows that different types of crimes behave differently over time. While some categories show steady growth, others exhibit fluctuations. This indicates that each category is influenced by distinct factors and requires separate attention for effective analysis and policy-making.

6.2.5 Relationship Analysis between Crime Categories

The relationship analysis suggests that there are observable patterns between different crime categories. Variations in one category may be associated with changes in another, indicating possible interdependencies. This highlights the need for multi-dimensional analysis when studying crime data.

6.2.6 Growth Rate Analysis

The growth rate analysis highlights periods of significant increase and decrease in crime levels. Certain years show noticeable spikes or declines, indicating major shifts in crime patterns. This provides a quantitative understanding of how crime evolves over time.

6.2.7 Distribution and Statistical Analysis

The statistical analysis reveals that crime data is unevenly distributed, with the presence of outliers and skewness. Some values are significantly higher than others, which affects the overall distribution. This emphasizes the importance of using appropriate techniques such as log transformation for better interpretation.

6.2.8 Correlation Analysis

The correlation analysis shows varying degrees of relationships between different variables. Some variables exhibit moderate relationships, while others show weaker connections. This indicates that multiple factors influence crime patterns, and no single variable fully explains the variations in the dataset.

6.3 Overall Insights

Based on the analyses performed, it can be concluded that crime patterns in India are complex and multi-dimensional. The data shows variations across time, regions, and categories, highlighting the need for comprehensive analysis.

The results emphasize that crime cannot be understood through a single perspective. Instead, a combination of trend analysis, category comparison, and statistical techniques is required to derive meaningful insights.

The interactive dashboard developed in this project plays a crucial role in enabling such multi-dimensional analysis by allowing users to explore data dynamically and gain deeper understanding of crime patterns.

6.4 Conclusion of Analysis

The analyses performed in this project successfully demonstrate the application of data science techniques in understanding real-world problems. By combining data preprocessing, visualization, and statistical methods, the project provides a structured approach to analyzing crime data.

The results obtained from the analysis highlight key patterns and trends that can be useful for further research, policy-making, and decision-making processes.

7. References

The following sources were directly used in this project for data collection, analysis, and implementation:

1. National Crime Records Bureau (NCRB), Government of India
   Crime in India Reports
   Available at: https://ncrb.gov.in

2. Open Government Data (OGD) Platform India
   Available at: https://data.gov.in

3. Pandas Documentation
   Available at: https://pandas.pydata.org/docs/

4. NumPy Documentation
   Available at: https://numpy.org/doc/

5. Plotly Documentation
   Available at: https://plotly.com/python/

6. Panel Documentation
   Available at: https://panel.holoviz.org/

7. hvPlot Documentation
   Available at: https://hvplot.holoviz.org/

8. Bibliography

The following resources were referred to during the development of this project for understanding concepts, tools, and implementation techniques:

1. Python Data Science Handbook – Jake VanderPlas

2. Online tutorials and documentation related to data visualization and dashboard development

3. Articles and blogs on crime data analysis and visualization techniques

4. YouTube tutorials on Panel, Plotly, and data analytics concepts

5. Various online coding forums such as Stack Overflow for troubleshooting and implementation support

9. Project Links

Google Drive (Project Files):
https://drive.google.com/drive/folders/1_luJY1cRhsJfPHNjj94yAPo0U08MmLf2?usp=sharingL

LinkedIn Post:
https://www.linkedin.com/posts/varnika-sharma-219525322_datascience-python-dataanalytics-ugcPost-7453399291122962433-3SBF?utm_source=social_share_send&utm_medium=member_desktop_web&rcm=ACoAAFGXYPUB7BFz7RpXoDOqfxjh-5KD4fwdkRY

GitHub Repository:

https://github.com/VarnikaSharma2176/Crime-Analysis